

Wind Turbine Foundations – Design Standard

Engineering Standard, Revision 3

Sofie Lindqvist · 20 February 2026 · WE-ENG-STD-001

1. Scope

This standard sets out Weyland's minimum requirements for the design of wind turbine support structures – offshore monopiles with transition pieces and onshore gravity foundations. It applies to all projects from concept design through to detailed design and to all designers working on our behalf, including foundation suppliers such as Rhenus Steelworks Foundations GmbH under WE-CON-2025-027.

Where a project-specific design basis exists (for Harrow Bay: HB-ENG-SPC-0042 Monopile Design Basis), the project document takes precedence for site-specific parameters. This standard governs everything else.

2. Normative references

- DNV-ST-0126 Support structures for wind turbines (current edition)
- DNV-ST-0437 Loads and site conditions for wind turbines
- IEC 61400-1 and IEC 61400-3-1 Design requirements
- EN 1993-1-9 Fatigue (with DNV-RP-C203 S-N curves for offshore details)
- ISO 19901-4 Geotechnical and foundation design considerations

3. Design principles

- Design life: 30 years minimum for all primary steel; 35 years where the project business case assumes life extension.
- Integrated load analysis (turbine + support structure) is mandatory for detailed design. Superelement exchange with the turbine supplier follows the interface protocol agreed under WE-CON-2025-022.
- Natural frequency of the first fore-aft mode shall lie in the soft-stiff range with at least 5 % margin to the 1P and 3P excitation bands, evaluated for best-estimate, lower- and upper-bound soil stiffness.
- Scour: either scour protection is designed in, or the monopile is designed for the full equilibrium scour depth (typically $1.3 \times D$) – never neither.
- Secondary steel (boat landings, J-tubes, platforms) is designed by the same designer of record as the primary steel.

4. Monopile geometry and fabrication

Parameter	Requirement	Note
Diameter to wall thickness (D/t)	≤ 110 in pile toe region	Local buckling during driving
Minimum wall thickness	50 mm	Handling and corrosion allowance
Corrosion allowance (splash zone)	6 mm + coating system	If no ICCP in splash zone
Out-of-roundness	≤ 0.5 % of D	Measured at each can

Pile tip	Flat, no driving shoe	Unless drivability study shows otherwise
Verticality after installation	$\leq 0.25^\circ$	Measured at pile head

Table 1 – Monopile geometric requirements

5. Geotechnical design

Lateral pile response shall be modelled with site-specific p-y curves calibrated against PISA-type 3D finite element analysis. API-type curves are only acceptable for concept design. Cyclic degradation is assessed for the storm load history defined in the design basis. Every foundation position shall have at least one CPT to the design penetration plus 5 m; one borehole per soil province is the minimum.

Lesson from Harrow Bay (LL-0024): early CPTs at three positions in the north-east corner stopped short of design depth on dense sand. Plan contingency CPT time in the survey contract.

6. Design verification

All foundation designs are subject to third-party certification (component certificate) and an internal design review at 30 %, 60 % and 90 % completion. Sign-off at each gate requires the Head of Engineering or a delegated chartered engineer.

7. Document control

Owner: Head of Engineering. Review cycle: every two years or after a major lesson learned.